CHL5233: Exploratory Data Analysis - Merged_Data

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The goal of this report is to preform the preliminary data analysis in preparation for the panel regression we will be conducting. We are examining the association between armed conflict and several mortality outcomes.

EXPOSURE = Armed Conflict (Binary – Y(1), N(0)) OUTCOME(S) = Rates per 100,000 live births (Continuous) 1. Maternal Mortality 2. Neonatal Mortality 3. Infant Mortality 4. Under 5 Mortality

We will begin by examining the variable types, names, and summary statistics.

```
raw_data <- read.csv("C:/Users/iamvi/OneDrive - University of Toronto/Year 2/Computation for Heal
    clean_names()

head(raw_data)
```

```
  iso year maternal_mortality neonatal_mortality infant_mortality
1 AFG 2000               1450               60.9             90.5
2 AFG 2001               1390               59.7             87.9
3 AFG 2002               1300               58.5             85.3
4 AFG 2003               1240               57.2             82.7
5 AFG 2004               1180               55.9             80.0
6 AFG 2005               1140               54.6             77.3
  under5_mortality earthquake drought total_conflicts armed_conflict  gdp_1000
1            129.2          0       1            5065              1        NA
2            125.2          1       0            5394              1        NA
3            121.1          1       0            5553              1 0.1835328
4            116.9          1       0            1157              1 0.2004626
5            112.6          1       0             944              1 0.2216576
6            108.4          1       0             817              1 0.2550551
  oecd pop_dens    urban  age_dep male_edu     temp rainfall_1000
1    0 14.13654 16.25324 108.3466 2.762086 12.69959     0.2763704
2    0 14.23156 16.25661 108.9899 2.856936 12.85570     0.2793079
```

```
3    0 14.32270 16.42654 109.3472 2.954241 12.71081     0.3805710
4    0 14.40691 16.60701 109.4475 3.054121 12.16592     0.4288939
5    0 15.21947 16.71367 109.2868 3.156706 13.04643     0.3754336
6    0 15.33619 16.85096 107.9646 3.262133 12.23141     0.4415680
```

```
tail(raw_data)
```

```
     iso year maternal_mortality neonatal_mortality infant_mortality
3715 ZWE 2014                494               28.2             42.9
3716 ZWE 2015                480               27.8             42.1
3717 ZWE 2016                468               27.4             40.8
3718 ZWE 2017                458               27.0             39.9
3719 ZWE 2018                 NA               26.6             38.8
3720 ZWE 2019                 NA               26.2             38.1
     under5_mortality earthquake drought total_conflicts armed_conflict
3715             62.7         NA      NA               0              0
3716             61.3         NA      NA               0              0
3717             58.7         NA      NA               0              0
3718             57.0          0       1               0              0
3719             54.8         NA      NA               0              0
3720             54.2         NA      NA               4              0
     gdp_1000 oecd pop_dens    urban  age_dep male_edu     temp rainfall_1000
3715 1.407034    0 26.52884 24.40427 85.87550 8.679591 20.87651     0.6777257
3716 1.410329    0 26.54454 24.75233 85.08337 8.785078 21.45470     0.4490721
3717 1.421788    0 26.53811 25.02842 84.11222 8.889947 21.39290     0.4939246
3718 1.192107    0 26.49281 25.29333 83.10129 8.994252 20.85962     0.9533149
3719 2.269177    0 26.47943 25.53759 82.12335 9.098048 20.86041     0.9535655
3720 1.421869    0 26.46341 25.70572 81.20786 9.201384 20.86120     0.9538138
```

```
slice_sample(raw_data, n=6)
```

```
  iso year maternal_mortality neonatal_mortality infant_mortality
1 COL 2015                85               8.4             13.4
2 KGZ 2004                79              19.7             35.6
3 USA 2008                14               4.3              6.5
4 KIR 2019                NA              21.9             40.3
5 DNK 2012                 5               3.0              3.5
6 GAB 2019                NA              20.1             31.5
  under5_mortality earthquake drought total_conflicts armed_conflict  gdp_1000
1             15.5          1       1             156              1  6.228426
2             41.1         NA      NA               0              0  0.433235
3              7.6          1       0               0              0 48.570046
4             51.2         NA      NA               0              0  1.432179
```

```
5              4.1         NA      NA               0              0 58.507508
6             43.0         NA      NA               0              0  7.523862
  oecd pop_dens      urban  age_dep  male_edu      temp rainfall_1000
1    0 39.39995 43.1958749 45.99655  8.021277 21.514094     1.3014736
2    0 19.95199 23.7846976 61.22216 10.386939  6.281799     0.4814955
3    1 53.79191 58.1279223 48.82407 12.671384 13.776051     0.9864733
4    0 60.75712  0.1092044 65.94983  8.427627 25.314130     1.7442547
5    1 38.04528 41.3017262 54.28406 13.043320  8.445198     0.6971362
6    0 29.12681 59.0952048 68.05746  9.185853 25.769733     2.1361730
```

Now we can look at the variable types

```
glimpse(raw_data)
```

```
Rows: 3,720
Columns: 18
$ iso                <chr> "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "AFG", "A~
$ year               <int> 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 200~
$ maternal_mortality <int> 1450, 1390, 1300, 1240, 1180, 1140, 1120, 1090, 103~
$ neonatal_mortality <dbl> 60.9, 59.7, 58.5, 57.2, 55.9, 54.6, 53.2, 51.7, 50.~
$ infant_mortality   <dbl> 90.5, 87.9, 85.3, 82.7, 80.0, 77.3, 74.6, 71.9, 69.~
$ under5_mortality   <dbl> 129.2, 125.2, 121.1, 116.9, 112.6, 108.4, 104.1, 99~
$ earthquake         <int> 0, 1, 1, 1, 1, 1, 1, NA, 0, 1, 1, 0, 1, 1, NA, 1, N~
$ drought            <int> 1, 0, 0, 0, 0, 0, 1, NA, 1, 0, 0, 1, 0, 0, NA, 0, N~
$ total_conflicts    <int> 5065, 5394, 5553, 1157, 944, 817, 1711, 4982, 7020,~
$ armed_conflict     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ gdp_1000           <dbl> NA, NA, 0.1835328, 0.2004626, 0.2216576, 0.2550551,~
$ oecd               <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ pop_dens           <dbl> 14.136539, 14.231558, 14.322705, 14.406906, 15.2194~
$ urban              <dbl> 16.25324, 16.25661, 16.42654, 16.60701, 16.71367, 1~
$ age_dep            <dbl> 108.34663, 108.98989, 109.34716, 109.44753, 109.286~
$ male_edu           <dbl> 2.762086, 2.856936, 2.954241, 3.054121, 3.156706, 3~
$ temp               <dbl> 12.69959, 12.85570, 12.71081, 12.16592, 13.04643, 1~
$ rainfall_1000      <dbl> 0.2763704, 0.2793079, 0.3805710, 0.4288939, 0.37543~
```

We see that the mortality variables are dbl = numeric with decimal places. We see amed conflict is an integer.

Total conflicts is also an integer, same with OECD, and year.

The rest of the covariates are dbl –> which algin.

```
summary(raw_data)
```

```
       iso            year      maternal_mortality neonatal_mortality
Length   :3720   Min.   :2000   Min.   :   2.0     Min.   : 0.80
N.unique : 186   1st Qu.:2005   1st Qu.:  17.0     1st Qu.: 4.90
N.blank  :   0   Median :2010   Median :  66.0     Median :12.10
Min.nchar:   3   Mean   :2010   Mean   : 210.6     Mean   :16.18
Max.nchar:   3   3rd Qu.:2014   3rd Qu.: 299.8     3rd Qu.:25.32
                 Max.   :2019   Max.   :2480.0     Max.   :60.90
                                NAs    :426        NAs    :20
infant_mortality under5_mortality   earthquake        drought
Min.   :  1.60   Min.   :  2.00   Min.   :0.0000   Min.   :0.0000
1st Qu.:  7.60   1st Qu.:  9.00   1st Qu.:0.0000   1st Qu.:0.0000
Median : 18.90   Median : 22.20   Median :1.0000   Median :1.0000
Mean   : 28.90   Mean   : 40.50   Mean   :0.5272   Mean   :0.5527
3rd Qu.: 44.52   3rd Qu.: 61.33   3rd Qu.:1.0000   3rd Qu.:1.0000
Max.   :138.10   Max.   :224.90   Max.   :1.0000   Max.   :1.0000
NAs    :20       NAs    :20       NAs    :3132     NAs    :3132
total_conflicts   armed_conflict      gdp_1000            oecd
Min.   :    0.0   Min.   :0.0000   Min.   :  0.1105   Min.   :0.000
1st Qu.:    0.0   1st Qu.:0.0000   1st Qu.:  1.2383   1st Qu.:0.000
Median :    0.0   Median :0.0000   Median :  4.0719   Median :0.000
Mean   :  361.1   Mean   :0.1892   Mean   : 11.4917   Mean   :0.171
3rd Qu.:    2.0   3rd Qu.:0.0000   3rd Qu.: 13.1531   3rd Qu.:0.000
Max.   :78644.0   Max.   :1.0000   Max.   :123.6787   Max.   :1.000
                                   NAs    :62
   pop_dens         urban           age_dep          male_edu
Min.   : 0.00   Min.   : 0.1025   Min.   : 16.17   Min.   : 1.067
1st Qu.:14.79   1st Qu.:17.2872   1st Qu.: 47.94   1st Qu.: 5.904
Median :27.52   Median :30.2535   Median : 55.51   Median : 8.368
Mean   :30.57   Mean   :30.6948   Mean   : 61.94   Mean   : 8.258
3rd Qu.:40.72   3rd Qu.:41.6558   3rd Qu.: 77.11   3rd Qu.:10.849
Max.   :99.86   Max.   :93.4135   Max.   :111.48   Max.   :14.441
NAs    :20      NAs    :20                        NAs    :20
     temp        rainfall_1000
Min.   :-2.405   Min.   :0.01993
1st Qu.:12.928   1st Qu.:0.59146
Median :21.958   Median :1.01288
Mean   :19.625   Mean   :1.20216
3rd Qu.:25.869   3rd Qu.:1.68706
Max.   :29.676   Max.   :4.71081
NAs    :20       NAs    :20
```

We can summize the following: - Year is between 2009 to 2019 - Maternal mortality has a very high max (2480 deaths per 100 000) - There are NAs for infant under 5, and neonatal (20) - There are 426 NAs for maternal mortality - armed conflict has no NA values - There appears to be no 'abnormal' negative values for the covariates and such

#quick check for duplicated rows#

```
sum(duplicated(raw_data))
```

```
[1] 0
```

There are none, so we are good to go!

##**Distribution of Outcomes** We can create a histogram for this (for each mortality variable)

```
ggplot(raw_data, aes(x = maternal_mortality)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 426 rows containing non-finite outside the scale range
(`stat_bin()`).
```

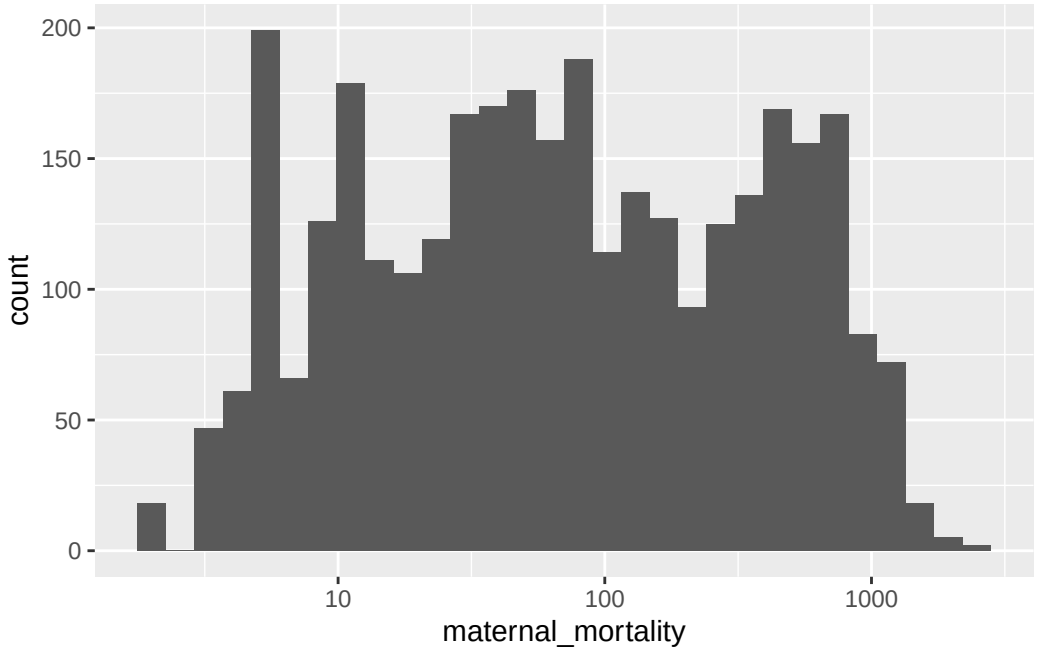


- We can see its very right skewed
- Very few "high" observations making it difficult to see majority of the observation values that lie in the lower range
- We can use log transform to look at **relative differences**. Shrink the scale to make it easier to examine.

```
ggplot(raw_data, aes(x = maternal_mortality)) +
  geom_histogram()+
  scale_x_log10()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 426 rows containing non-finite outside the scale range
(`stat_bin()`).
```

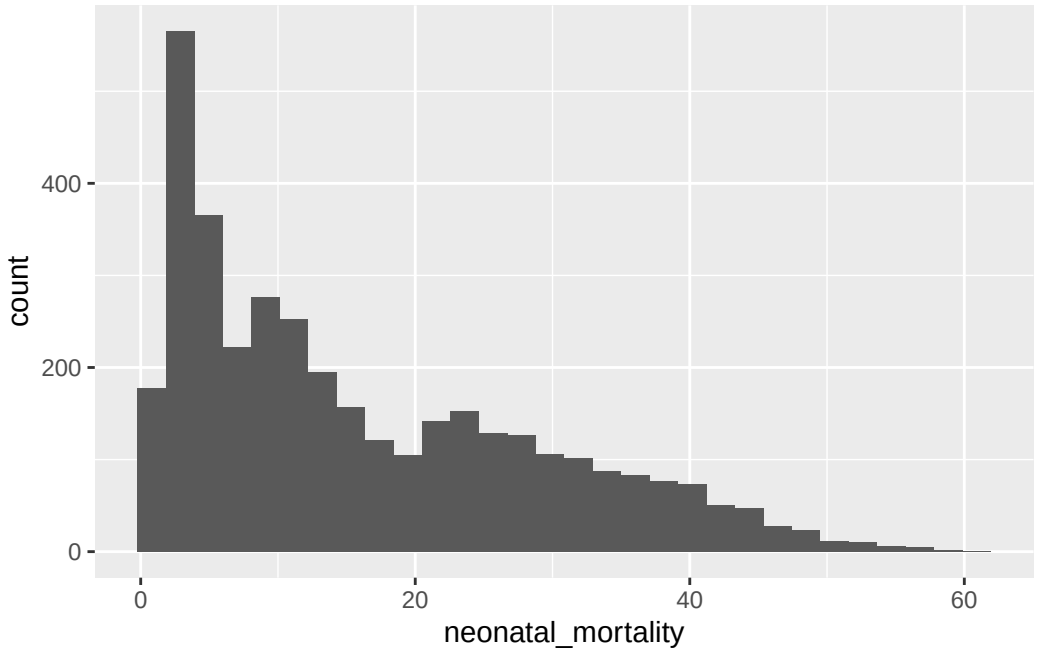


Clearly we can see that mortality rates are very right skewed as most concentrate in the lower value range. small number of observations at very high rates.

```
ggplot(raw_data, aes(x = neonatal_mortality)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

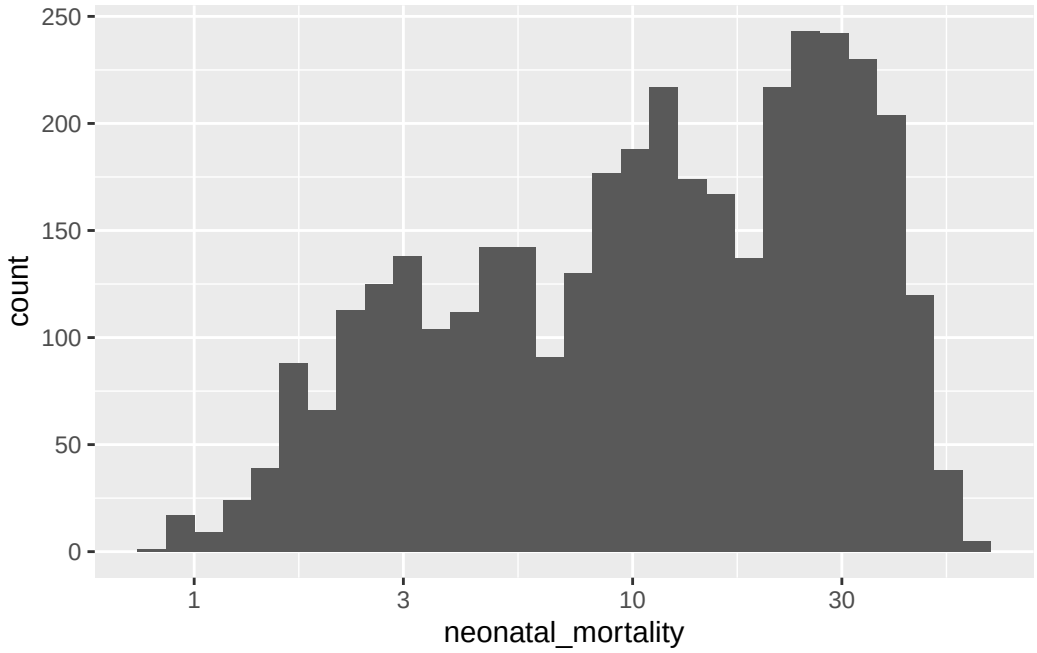
```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
ggplot(raw_data, aes(x = neonatal_mortality)) +
  geom_histogram()+
  scale_x_log10()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_bin()`).
```

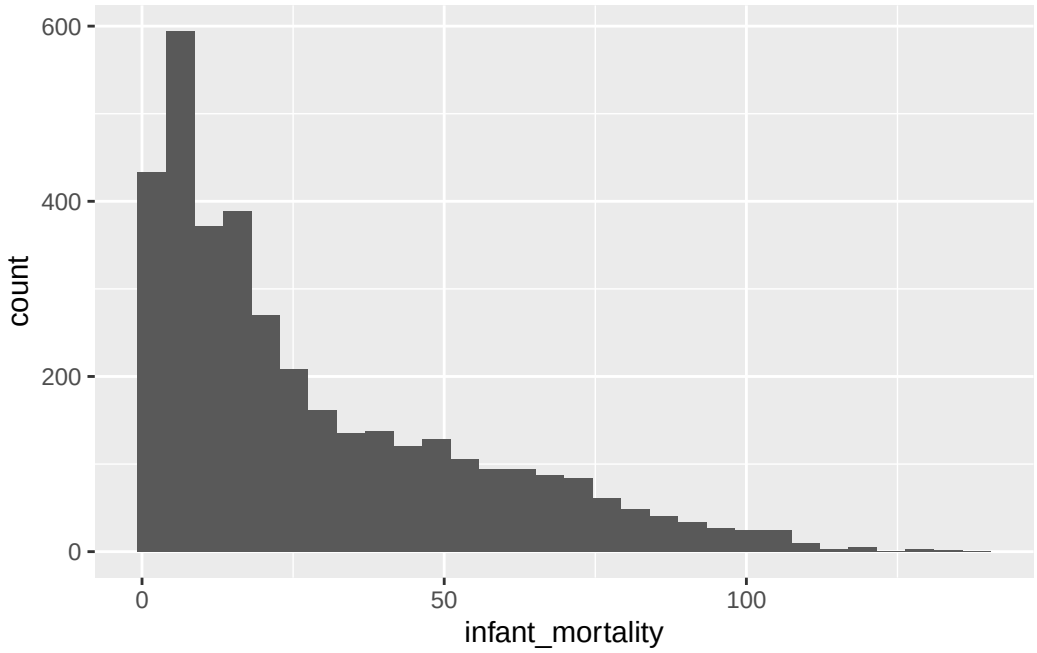


Here we can see a lot of neonatal mortality is on the slightly higher end value wise.

```
ggplot(raw_data, aes(x = infant_mortality)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

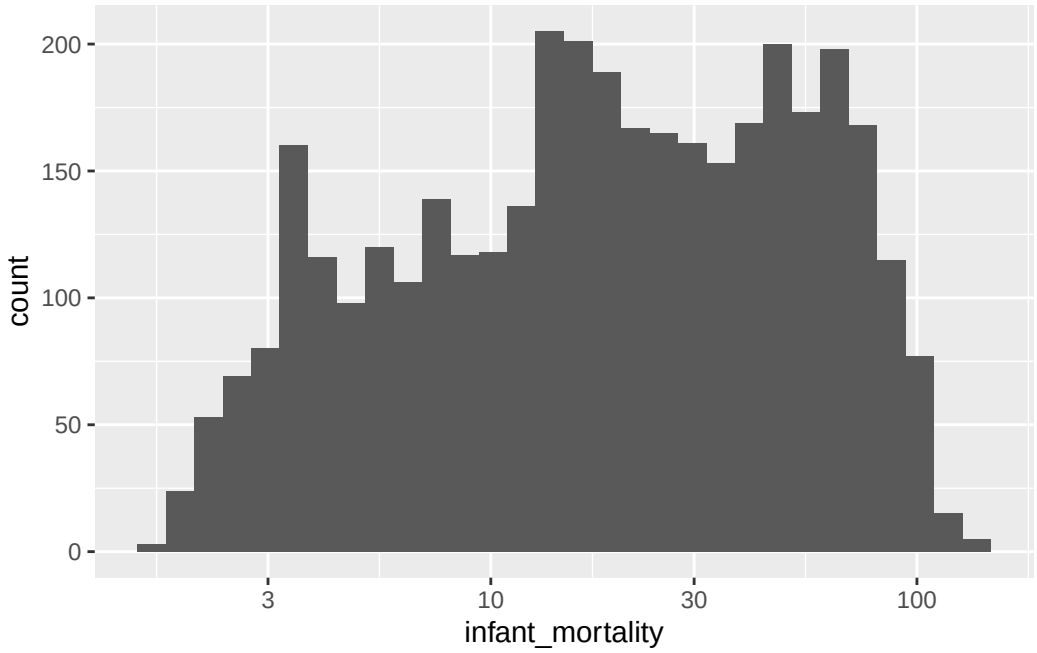
```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
ggplot(raw_data, aes(x = infant_mortality)) +
  geom_histogram()+
  scale_x_log10()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_bin()`).
```

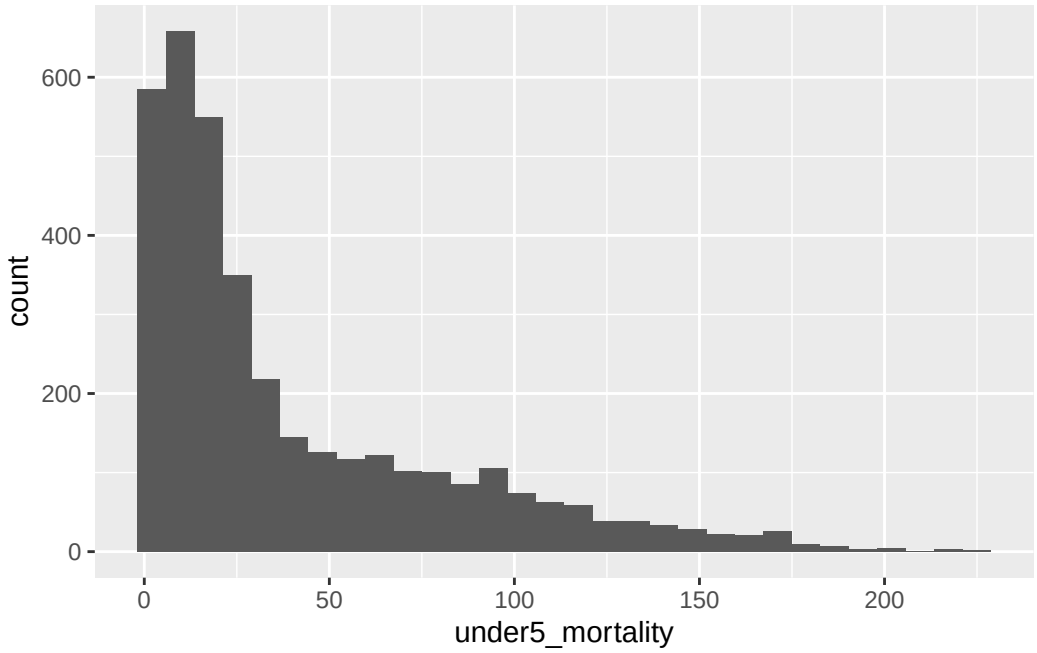


Similar to neonatal mortality

```
ggplot(raw_data, aes(x = under5_mortality)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

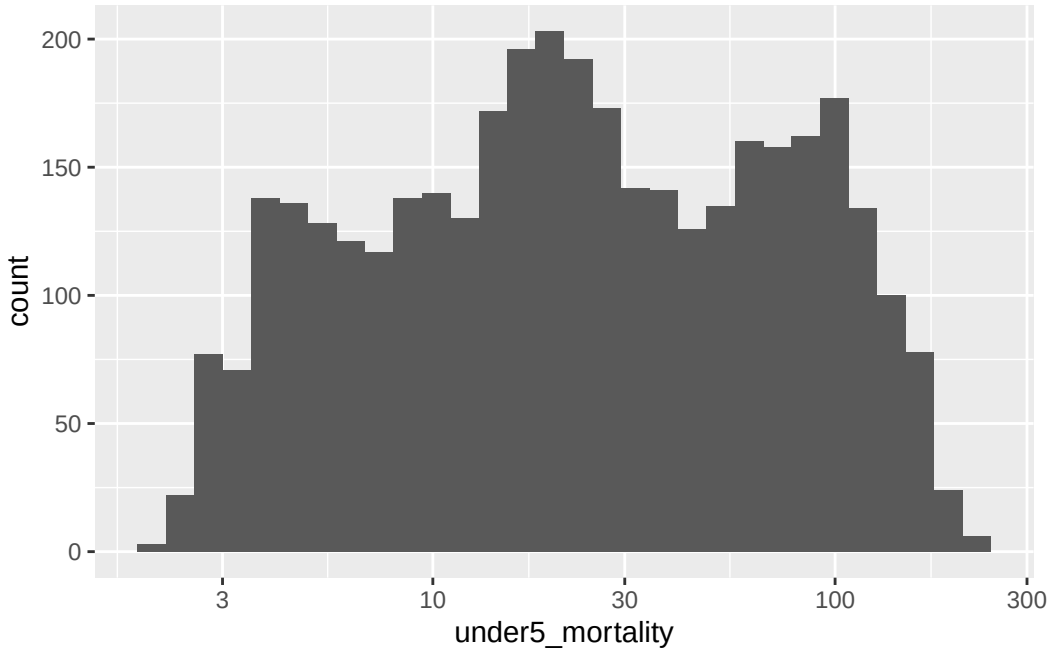
```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
ggplot(raw_data, aes(x = under5_mortality)) +
  geom_histogram()+
  scale_x_log10()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_bin()`).
```



Clearly we see maternal mortality is much lower compared to younger-onset mortality (infant, neonatal, under5)

##Examine maternal mortality

```
summary(raw_data$maternal_mortality)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
    2.0    17.0    66.0   210.6   299.8  2480.0     426
```

```
#lets examine the max observation
summary(raw_data$maternal_mortality)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
    2.0    17.0    66.0   210.6   299.8  2480.0     426
```

```
raw_data |>
  filter(maternal_mortality == max(maternal_mortality, na.rm = TRUE)) |>
  select(iso, year, maternal_mortality, armed_conflict, total_conflicts) |>
  kable(caption = "Country with Highest Maternal Mortality")
```

Table 1: Country with Highest Maternal Mortality

iso	year	maternal_mortality	armed_conflict	total_conflicts
SLE	2000	2480	1	4493

Based on that observation -> SLE (Sierra Leonne), had 4493 conflicts Let's confirm with the raw data. Seems to be the case. very consistently high across all years. This is why the data is super skewed.

Some background research –> Sierra Leonne civil war that occured during that time.

##EXAMINE FREQUENCY OF OUTCOME WITHIN EACH## Now we will explore the proportion of 'armed_conflict' overall

First we convert to a factor variable rather than numerical

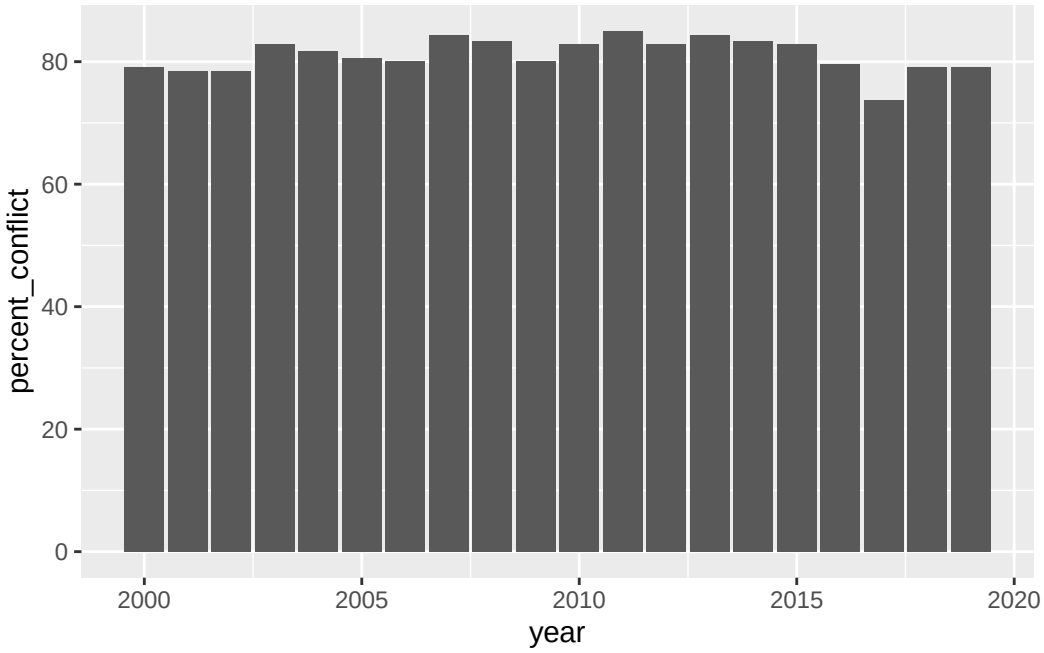
```
conflict_data <- raw_data |> mutate(armed_conflict=as.factor(armed_conflict))
summary(conflict_data$armed_conflict) #for counts
prop.table(table(conflict_data$armed_conflict)) #for proportions

#turn it back into numeric for purpose of regression fixed effects
conflict_data <- conflict_data |> mutate(armed_conflict=as.numeric(armed_conflict))
```

81 % of countries across all years did not experience an armed conflict 18.9% experienced an armed conflict This is overall so we can get the prop by year

```
yearly <- conflict_data %>% group_by(year) %>%
    summarize(percent_conflict = 100*mean(armed_conflict==1, na.rm=TRUE))
#na.rm = true just means include na observations (there aren't any i believe)


ggplot(yearly, aes(x = year, y = percent_conflict)) +
  geom_bar(stat = "identity") #identity to make bar height reflect the percentages in y.Can also
```



##MISSING DATA

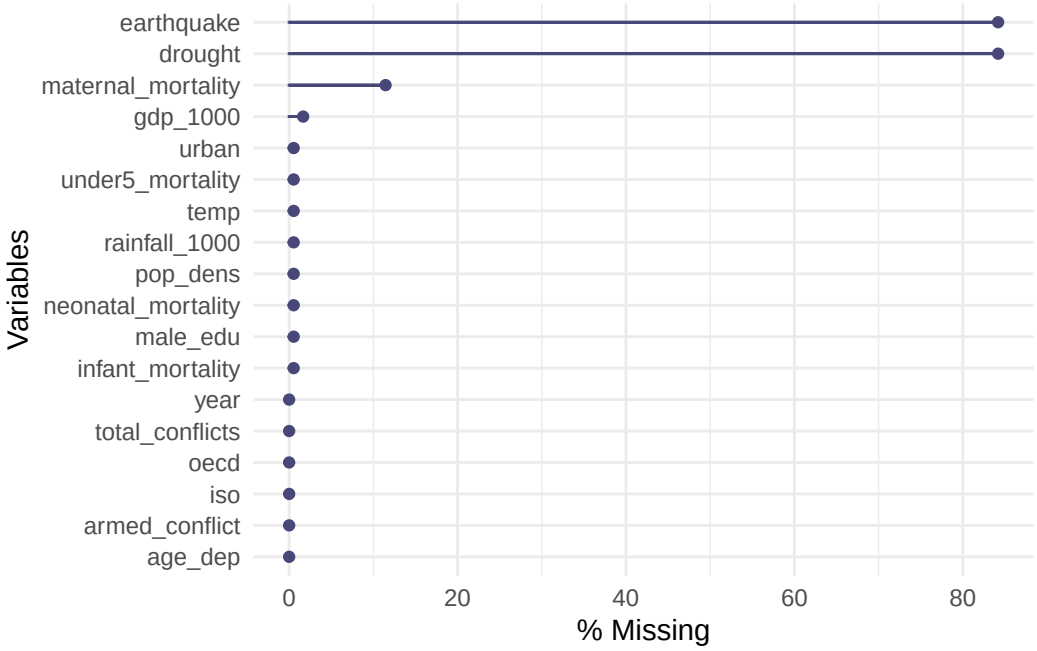
Now we need to examine the missingness within each variable. We will pay particular close attention to our exposure and outcomes of interest.

```
naniar::pct_miss_case(conflict_data)
```

```
[1] 85.99462
```

85% of cases have some sort of missing value

```
gg_miss_var(conflict_data, show_pct=TRUE)
```



Clearly we can see: - Earthquake and drought have high missingness counts - our exposure (armed conflict) seems to have no missing data (yay) - Some outcomes have missing data (maternal_mortalityu, under5, neonatal, infant)

Lets quantify the exact amount however for the exposure and outcomes.

```
missing_table <- data.frame(
  Variable = names(conflict_data),
  Missing_n = colSums(is.na(conflict_data)),
  Missing_pct = round(colMeans(is.na(conflict_data)) * 100, 1))

#format it in a nice table
kable(
  missing_table,
  col.names = c("Variable", "Missing (n)", "Missing (%)"),
  caption = "Missingness quanitified by variable"
)
```

Table 2: Missingness quanitified by variable

	Variable	Missing (n)	Missing (%)
iso	iso	0	0.0
year	year	0	0.0
maternal_mortality	maternal_mortality	426	11.5
neonatal_mortality	neonatal_mortality	20	0.5

	Variable	Missing (n)	Missing (%)
infant_mortality	infant_mortality	20	0.5
under5_mortality	under5_mortality	20	0.5
earthquake	earthquake	3132	84.2
drought	drought	3132	84.2
total_conflicts	total_conflicts	0	0.0
armed_conflict	armed_conflict	0	0.0
gdp_1000	gdp_1000	62	1.7
oecd	oecd	0	0.0
pop_dens	pop_dens	20	0.5
urban	urban	20	0.5
age_dep	age_dep	0	0.0
male_edu	male_edu	20	0.5
temp	temp	20	0.5
rainfall_1000	rainfall_1000	20	0.5

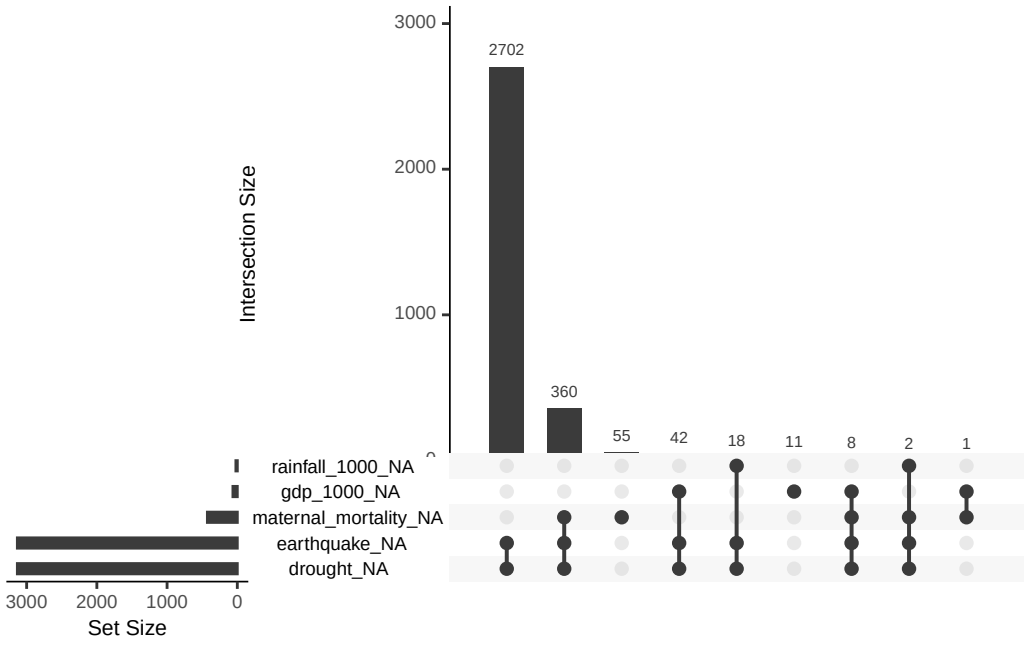
We should explore the earthquake and drought, by exposure and outcome to determine the reason for missingness (MNAR, MAR, MCAR)

```
gg_miss_upset(conflict_data)
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the UpSetR package.
  Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the UpSetR package.
  Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.

Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
i Please use the `linewidth` argument instead.
i The deprecated feature was likely used in the UpSetR package.
  Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.
```

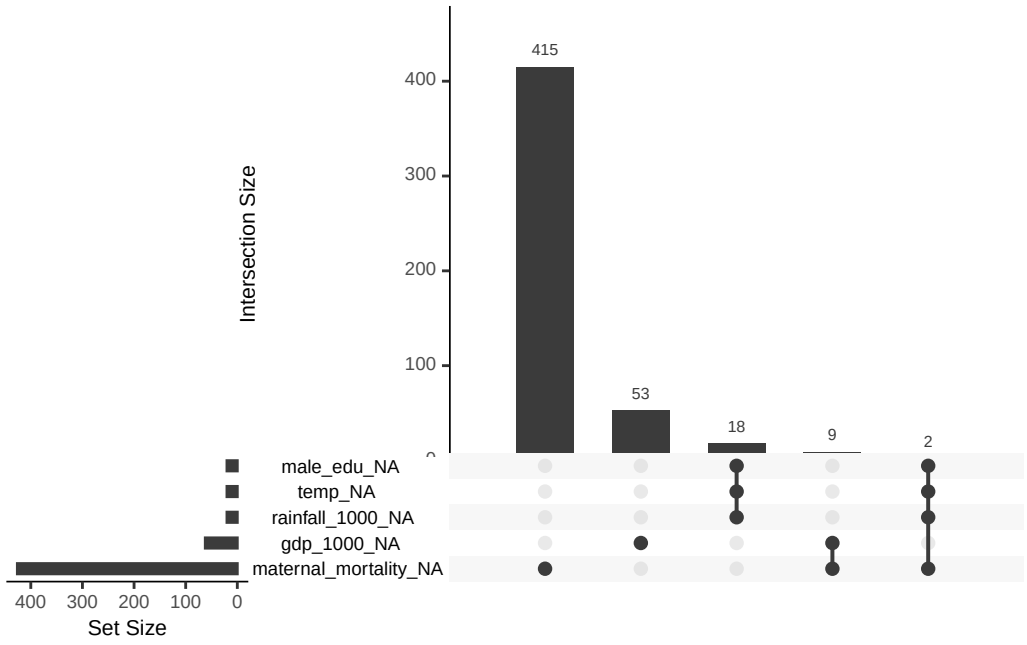


This upset plot helps us see whether variables tend to be missing together. Earthquake and drought seem to be. Set size = number of missing observations. Interaction size is how many they have in common.

This plot seems to suggest that earthquake and drought are missing together, which may indicate we should explore why.

This is likely actually due to the way we had coded the earthquake and drought variables. Since when we merged all the final data sets, any countries that had no earthquake or drought, were just coded as NA.

```
conflict_data_recode <- conflict_data %>%
  mutate(
    earthquake = replace_na(earthquake, 0),
    drought = replace_na(drought, 0)
  )
gg_miss_upset(conflict_data_recode)
```



Now for maternal mortality, let's try to explore the reasons for missingness. Seem to potentially not have that data available. Lets look at these observations closely:

```
conflict_data_recode %>%
  filter(is.na(maternal_mortality)) %>%
  count(iso, sort = TRUE)
```

```
    iso  n
1   AND 20
2   DMA 20
3   MHL 20
4   AFG  2
5   AGO  2
6   ALB  2
7   ARE  2
8   ARG  2
9   ARM  2
10  ATG  2
11  AUS  2
12  AUT  2
13  AZE  2
14  BDI  2
15  BEL  2
16  BEN  2
17  BFA  2
```

```
18  BGD  2
19  BGR  2
20  BHR  2
21  BHS  2
22  BIH  2
23  BLR  2
24  BLZ  2
25  BOL  2
26  BRA  2
27  BRB  2
28  BRN  2
29  BTN  2
30  BWA  2
31  CAF  2
32  CAN  2
33  CHE  2
34  CHL  2
35  CHN  2
36  CIV  2
37  CMR  2
38  COD  2
39  COG  2
40  COL  2
41  COM  2
42  CPV  2
43  CRI  2
44  CUB  2
45  CYP  2
46  CZE  2
47  DEU  2
48  DJI  2
49  DNK  2
50  DOM  2
51  DZA  2
52  ECU  2
53  EGY  2
54  ERI  2
55  ESP  2
56  EST  2
57  ETH  2
58  FIN  2
59  FJI  2
60  FRA  2
61  FSM  2
62  GAB  2
```

```
63  GBR  2
64  GEO  2
65  GHA  2
66  GIN  2
67  GMB  2
68  GNB  2
69  GNQ  2
70  GRC  2
71  GRD  2
72  GTM  2
73  GUY  2
74  HND  2
75  HRV  2
76  HTI  2
77  HUN  2
78  IDN  2
79  IND  2
80  IRL  2
81  IRN  2
82  IRQ  2
83  ISL  2
84  ITA  2
85  JAM  2
86  JOR  2
87  JPN  2
88  KAZ  2
89  KEN  2
90  KGZ  2
91  KHM  2
92  KIR  2
93  KOR  2
94  KWT  2
95  LAO  2
96  LBN  2
97  LBR  2
98  LBY  2
99  LCA  2
100 LKA  2
101 LSO  2
102 LTU  2
103 LUX  2
104 LVA  2
105 MAR  2
106 MDA  2
107 MDG  2
```

```
108 MDV  2
109 MEX  2
110 MKD  2
111 MLI  2
112 MLT  2
113 MMR  2
114 MNE  2
115 MNG  2
116 MOZ  2
117 MRT  2
118 MUS  2
119 MWI  2
120 MYS  2
121 NAM  2
122 NER  2
123 NGA  2
124 NIC  2
125 NLD  2
126 NOR  2
127 NPL  2
128 NZL  2
129 OMN  2
130 PAK  2
131 PAN  2
132 PER  2
133 PHL  2
134 PNG  2
135 POL  2
136 PRI  2
137 PRK  2
138 PRT  2
139 PRY  2
140 QAT  2
141 ROU  2
142 RUS  2
143 RWA  2
144 SAU  2
145 SDN  2
146 SEN  2
147 SGP  2
148 SLB  2
149 SLE  2
150 SLV  2
151 SOM  2
152 SRB  2
```

```
153 SSD  2
154 STP  2
155 SUR  2
156 SVK  2
157 SVN  2
158 SWE  2
159 SWZ  2
160 SYC  2
161 SYR  2
162 TCD  2
163 TGO  2
164 THA  2
165 TJK  2
166 TKM  2
167 TLS  2
168 TON  2
169 TTO  2
170 TUN  2
171 TUR  2
172 TZA  2
173 UGA  2
174 UKR  2
175 URY  2
176 USA  2
177 UZB  2
178 VCT  2
179 VEN  2
180 VNM  2
181 VUT  2
182 WSM  2
183 YEM  2
184 ZAF  2
185 ZMB  2
186 ZWE  2
```

So this tells us countries with AND, DMA, and MHL iso codes have 20 missing values. DMA = Dominica MHL = Marshall Islands AND = Andorra

```
conflict_data_recode %>%
  filter(is.na(maternal_mortality)) %>%
  count(year, sort = TRUE)
```

```
   year   n
1  2018 186
```

```
2  2019 186
3  2000   3
4  2001   3
5  2002   3
6  2003   3
7  2004   3
8  2005   3
9  2006   3
10 2007   3
11 2008   3
12 2009   3
13 2010   3
14 2011   3
15 2012   3
16 2013   3
17 2014   3
18 2015   3
19 2016   3
20 2017   3
```

Majority in 2018 and 2019.

Overall we can say: *Missing maternal mortality data were concentrated in specific countries (Andorra, Dominica, and Marshall Islands) and primarily occurred in 2018 and 2019. Because the missingness appears related to observed characteristics of the data (country and year), it is unlikely to be Missing Completely At Random (MCAR) and is more consistent with a Missing At Random (MAR). We will likely retain these covariates regardless.*

##Dealing with Missing Data!

We will use a complete case analysis approach to remove observations that are missing any of the outcomes and exposures. **Do not do this, will mess up regression, ignore**

Therefore we can see a total of 444 observations were removed. We will likely use sensitivity analyses later on to check this decision.

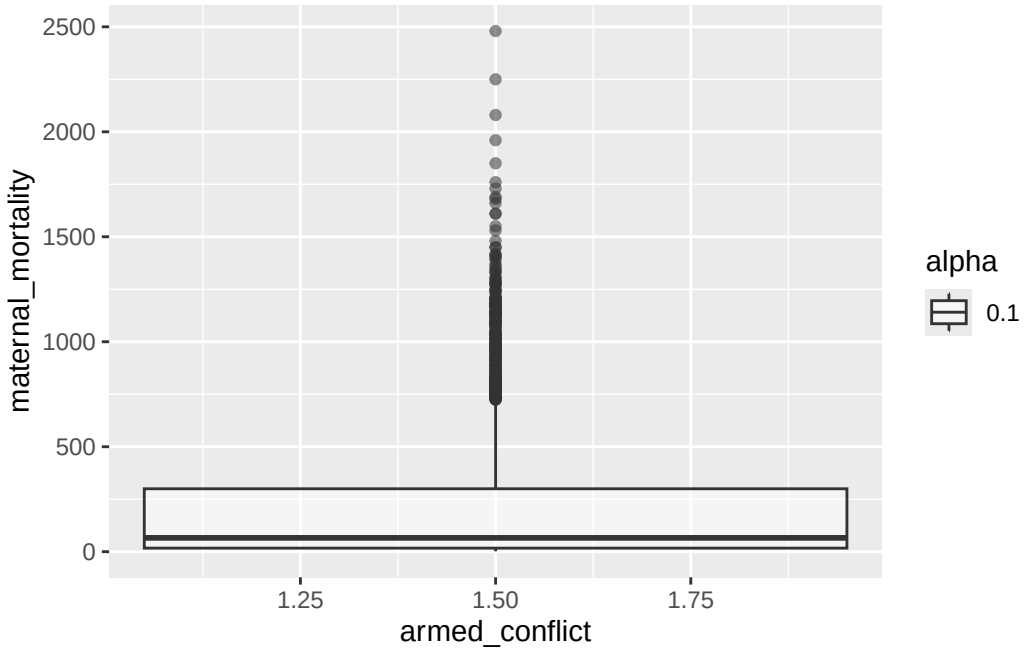
##Understanding relationships between variables##

```
conflict_data |>
  ggplot(aes(x = armed_conflict, y = maternal_mortality, alpha = 0.1)) +
  geom_boxplot()
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 426 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



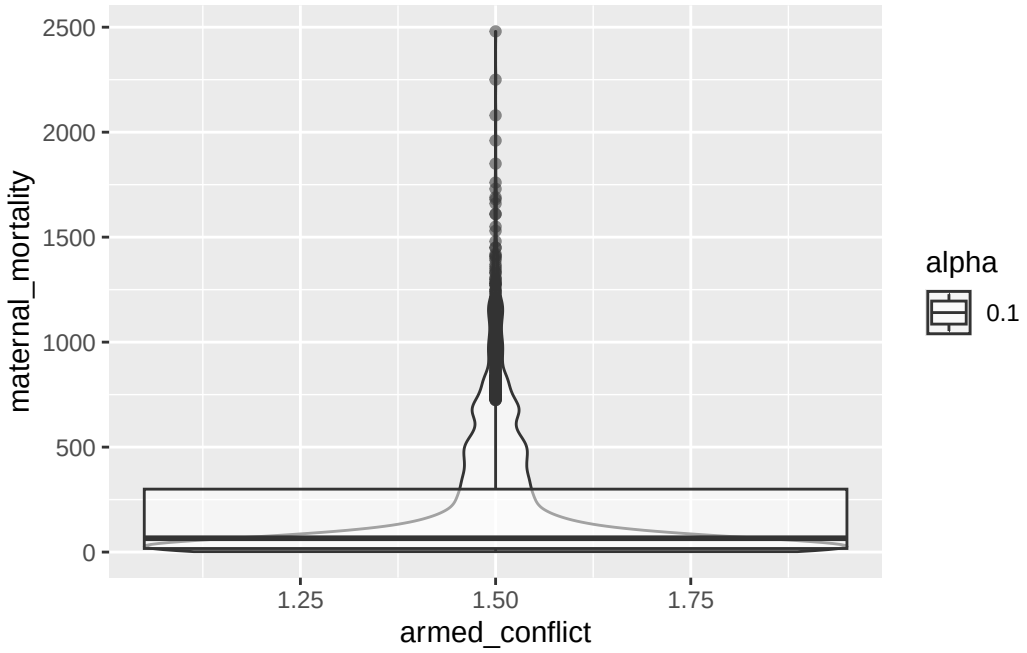

```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = maternal_mortality, alpha = 0.1)) +
  geom_violin()+
  geom_boxplot(width=0.1)
```

```
Warning: Removed 426 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 426 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



We do see the median maternal mortality is lower among no armed conflict, compared to armed conflict. Suggest country years with armed conflict have higher mortatlity (maternal)

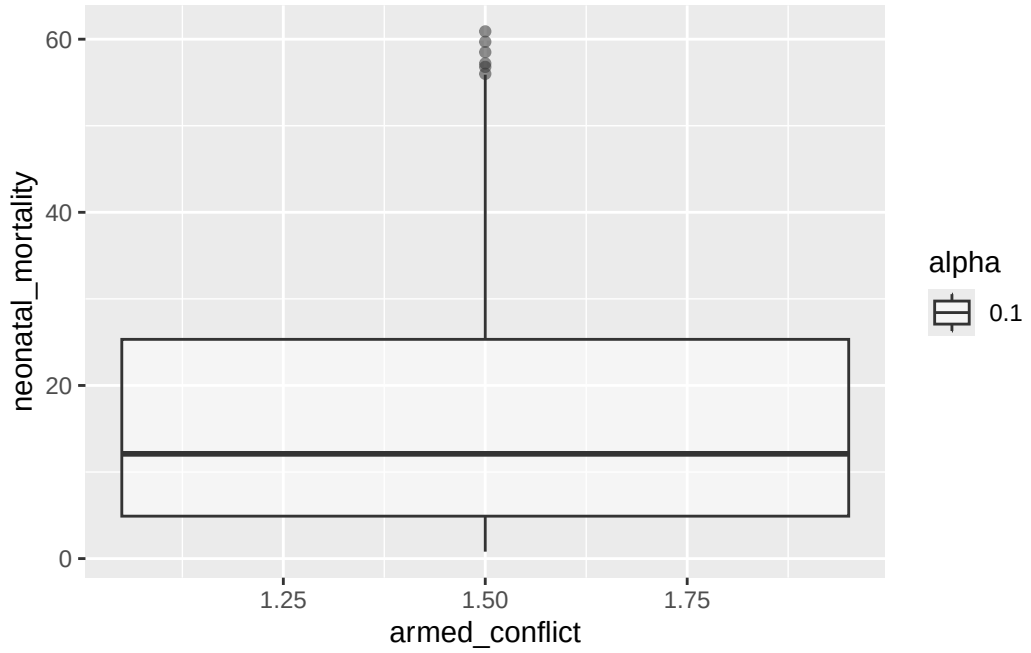
Conflict group has a greater spread (values extend high). Seem to be very skewed (most groups are on the lowe rend but there is a tail on the higher end)

```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = neonatal_mortality, alpha = 0.1)) +
  geom_boxplot()
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



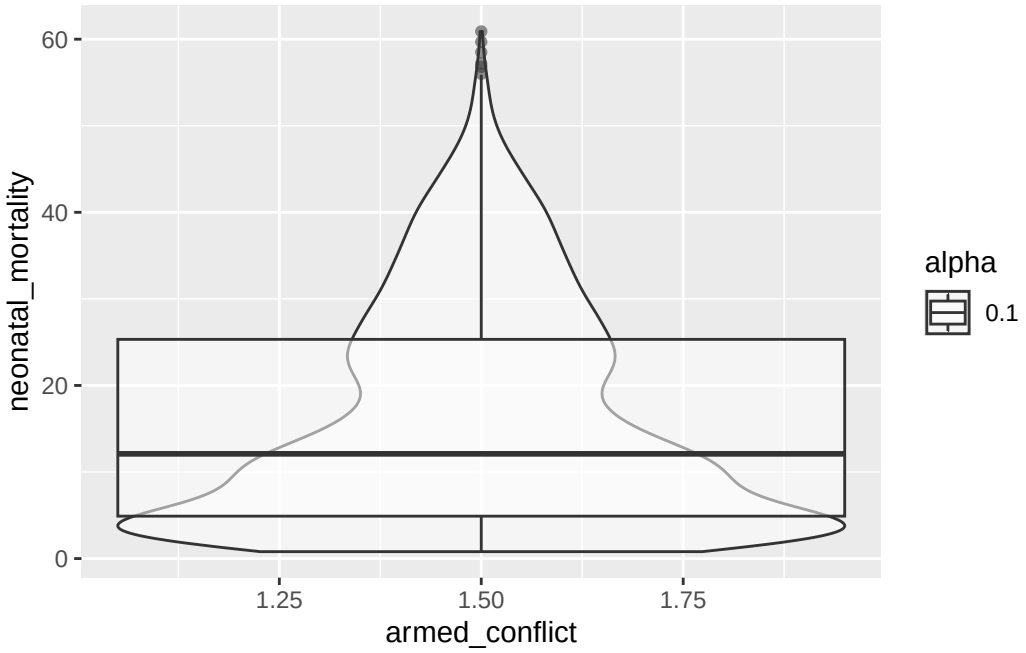

```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = neonatal_mortality, alpha = 0.1)) +
  geom_violin()+
  geom_boxplot(width=0.1)
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



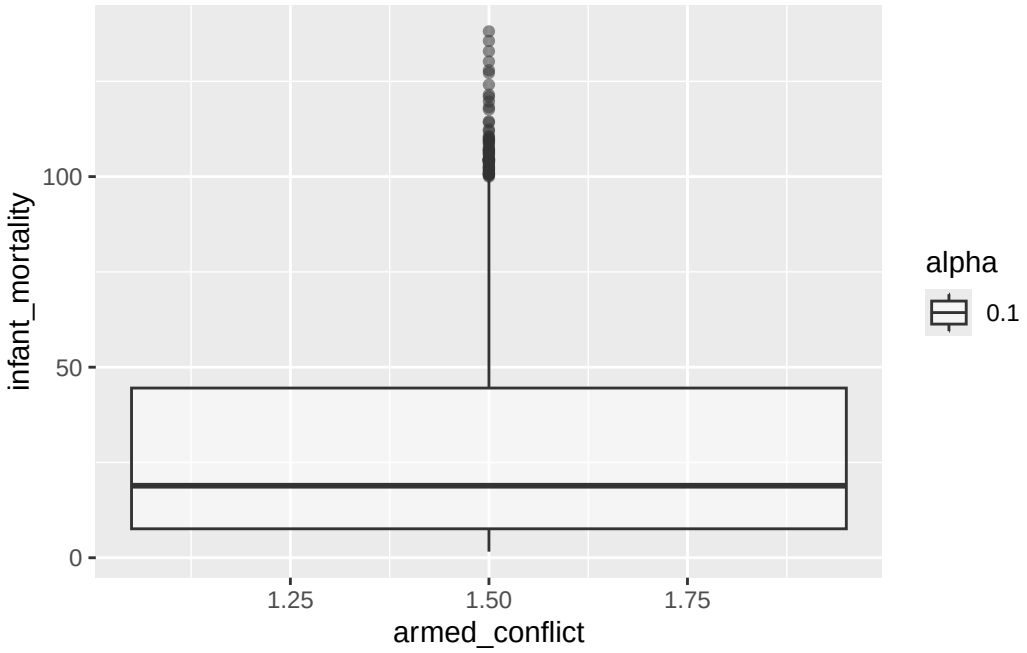
Similar trends for neonatal in terms of median in armed 0 vs 1. However much more higher values, not as skewed.

```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = infant_mortality, alpha = 0.1)) +
  geom_boxplot()
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



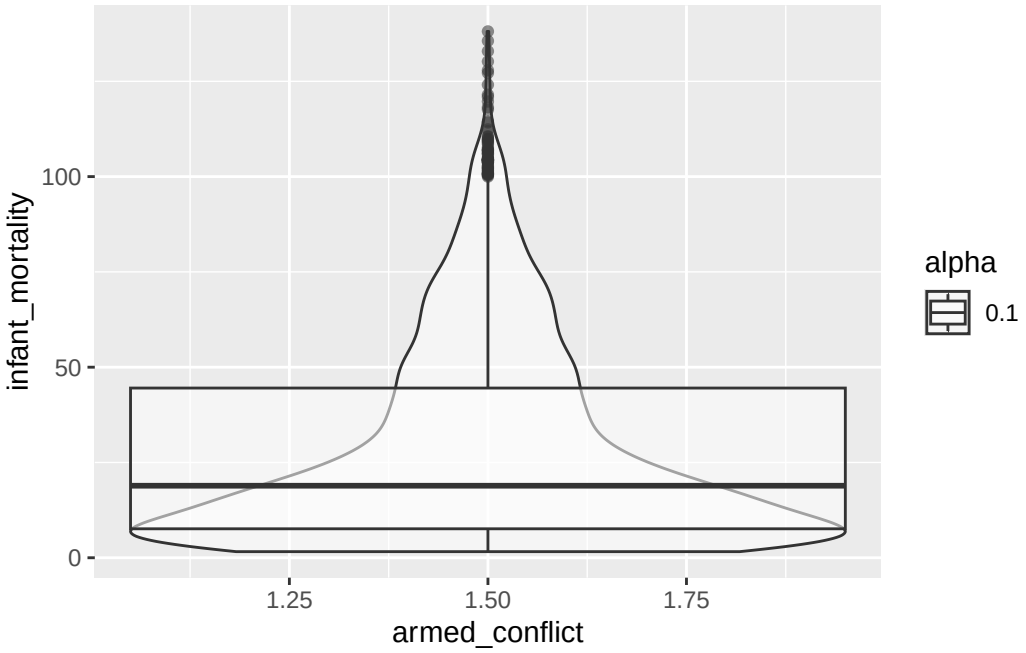

```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = infant_mortality, alpha = 0.1)) +
  geom_violin()+
  geom_boxplot(width=0.1)
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

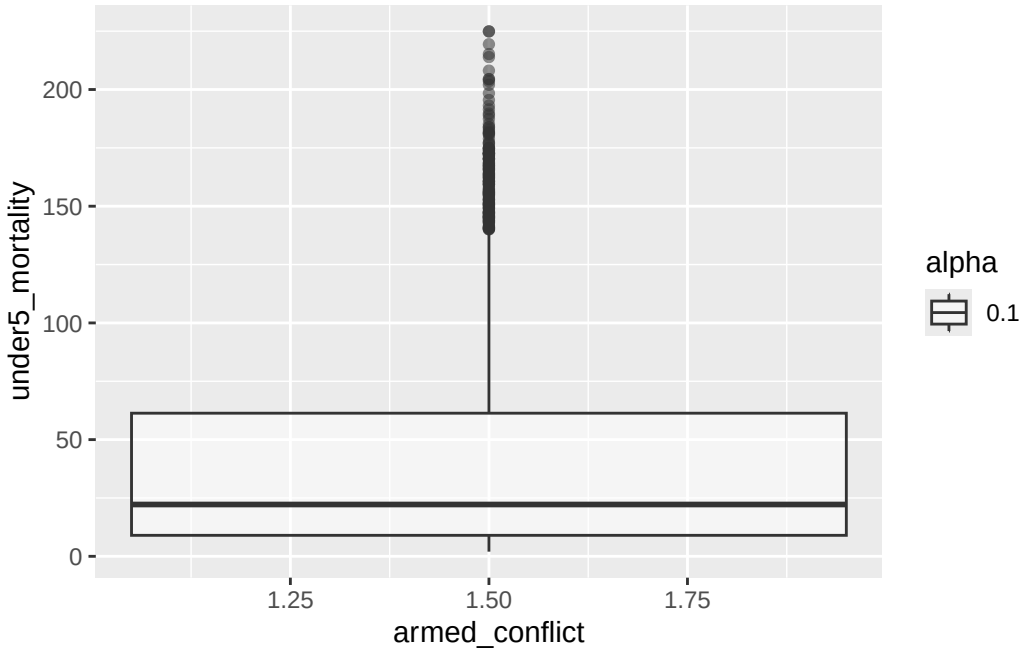



```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = under5_mortality, alpha = 0.1)) +
  geom_boxplot()
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



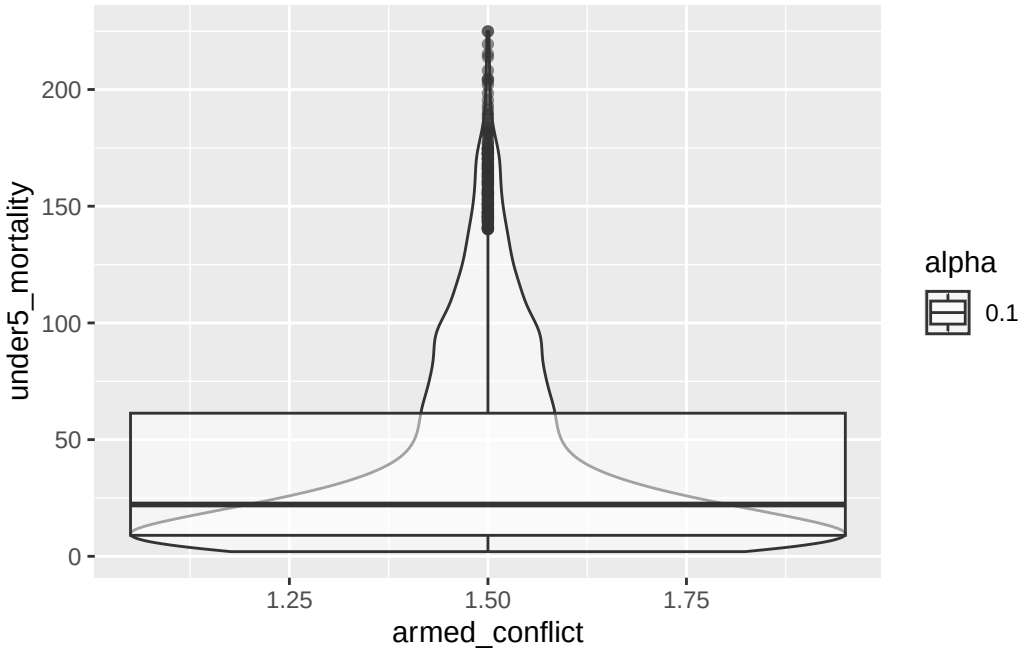

```
conflict_data|>
  ggplot(aes(x = armed_conflict, y = under5_mortality, alpha = 0.1)) +
  geom_violin()+
  geom_boxplot(width=0.1)
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

```
Warning: Orientation is not uniquely specified when both the x and y aesthetics are
continuous. Picking default orientation 'x'.
```

```
Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?
```

```
Warning: Removed 20 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



Save the conflict df into a rds file! This saves data for future use in R format. We will use this for the next in class assignment.

```
library(here)
```

```
here() starts at C:/Users/iamvi/OneDrive - University of Toronto/Year 2/Computation for Health Da
```

```
saveRDS(
  conflict_data,
  here("data", "processed", "df_clean.rds")
)
```